Chapter 3: The Cox Regression Model

Modelling Survival Data in Medical Research

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Part 1

Chapter 3 Outline

1. Why Go Beyond Chapter 2?
2. The Proportional Hazards Model
3. Linear Component: Variates & Factors
4. Fitting via Partial Likelihood
5. Confidence Intervals & Wald Tests

Where We Are and Where We Are Going

In Chapter 2, we learned powerful **non-parametric tools**:

- Kaplan-Meier estimate of the survivor function $\hat{S}(t)$
- Greenwood's formula for standard errors
- Median and percentiles of survival times
- Log-rank, Wilcoxon, Peto-Peto tests to compare two groups

The fundamental limitation

All of Chapter 2's methods handle **one group variable at a time**.
Real studies record dozens of variables per patient. We cannot study them one at a time.

Chapter 3 introduces the **Cox regression model** — a framework that handles many variables simultaneously, estimates their effects, and extends the log-rank test.

A Concrete Scenario: Multiple Myeloma

A researcher has data on 48 patients with **multiple myeloma**, including:

Variable	Type
Age	Continuous
Sex	Binary (M/F)
Bun (blood urea nitrogen)	Continuous
Ca (serum calcium)	Continuous
Hb (serum haemoglobin)	Continuous
Pcells (% plasma cells)	Continuous
Protein (Bence-Jones)	Binary

Research questions:

- Does higher Bun increase the risk of death?
- After adjusting for Bun, does Hb still matter?
- Which combination of variables best predicts survival?
- How do we estimate survival for a new patient with specific values?

The log-rank test could only compare, say, high-Bun vs. low-Bun patients. It cannot adjust for all other variables at once.

The Statistical Modelling Approach

Two reasons to model survival data

1. **Inference:** determine *which* variables affect the hazard, and *by how much*.
2. **Prediction:** estimate $h(t)$, $S(t)$, and median survival for individuals with specific covariate values.

Why model the hazard $h(t)$?

- It is the natural measure of instantaneous risk
- Covariates enter multiplicatively (≥ 0 guaranteed)
- Directly connects to $S(t)$ and $H(t)$
- More tractable mathematically than modelling T directly

Semi-parametric = flexible

- We specify the *covariate effects* parametrically ($\boldsymbol{\beta}$)
- We leave the baseline hazard shape $h_0(t)$ completely *unspecified*
- No distributional assumption needed
- Robust to misspecification

Three Approaches to Survival Analysis

Non-Parametric
Chapter 2

No assumptions.
Kaplan-Meier, log-rank.
Cannot adjust
for covariates.

add covariates →

Semi-Parametric
Chapter 3 (Cox)

β estimated.
$h_0(t)$ unspecified.
Flexible and widely used.

specify $h_0(t)$ →

Parametric
Chapters 5–6

$h_0(t)$ specified
(Weibull, etc.).
Efficient but
assumption-dependent.

The Cox model is the most widely used model in medical survival analysis.
It builds directly on the ideas of Chapter 2 while overcoming its limitations.

Reviewing the Hazard Function

Before building the Cox model, recall from Chapter 1:

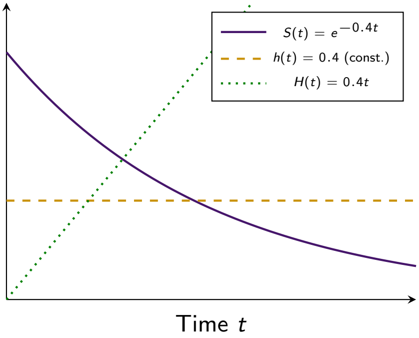
Hazard function

$$h(t) = \lim_{\delta t \to 0} \frac{P(t \leq T < t + \delta t \mid T \geq t)}{\delta t}$$

The instantaneous risk of death at t, given survival to t.

Three equivalent representations:

$$H(t) = \int_0^t h(u)\,du = -\log S(t), \quad S(t) = e^{-H(t)}$$



The Simple Two-Group Model

Suppose two groups: **standard treatment** (S) and **new treatment** (N).

Proportional hazards assumption

$$h_N(t) = \psi \, h_S(t) \quad \text{for all } t \geq 0$$

$\psi > 0$ is the **hazard ratio**. Write $\psi = e^{\beta}$ so β can take any real value.

Interpreting ψ:

- $\psi < 1$: new treatment reduces risk (better)
- $\psi = 1$: no difference
- $\psi > 1$: new treatment increases risk (worse)

Consequence for $S(t)$:

If $h_N(t) = \psi \, h_S(t)$, then integrating:

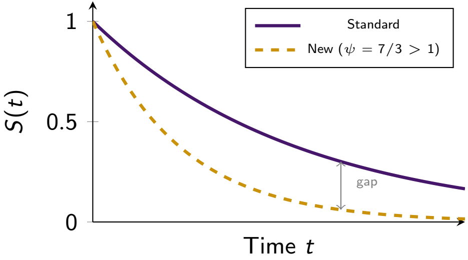
$$S_N(t) = [S_S(t)]^{\psi}$$

The two survivor functions **never cross**.

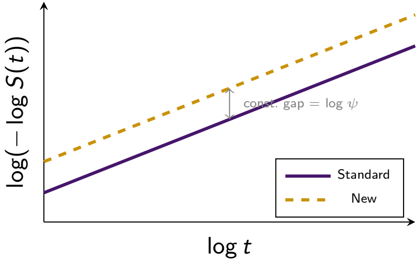
On a log scale, the gap is constant.

Proportional Hazards — What "Never Crossing" Looks Like

Original scale:



Log-log scale (straight lines if PH holds):



How to check PH informally: plot $\log(-\log \hat{S}(t))$ vs. $\log t$ for each group. If PH holds, the lines should be roughly *parallel*.

The General Cox Regression Model

Cox regression model

$$h_i(t) = \exp(\beta_1 x_{1i} + \cdots + \beta_p x_{pi})\ h_0(t)$$

- $h_0(t)$: **baseline hazard** — hazard when all $x_j = 0$ (shape unspecified)
- e^{β_j}: hazard ratio per 1-unit increase in X_j (to be estimated)

Equivalent log-linear form:

$$\log \frac{h_i(t)}{h_0(t)} = \underbrace{\beta_1 x_{1i} + \cdots + \beta_p x_{pi}}_{\eta_i \text{ (linear predictor)}}$$

The log-hazard ratio is *linear* in the covariates.

No constant term: any β_0 would rescale $h_0(t)$ and cancel.

Named after Sir David Cox (1972) — one of the most cited papers in statistics.

Semi-parametric: $\boldsymbol{\beta}$ is estimated without ever specifying $h_0(t)$.

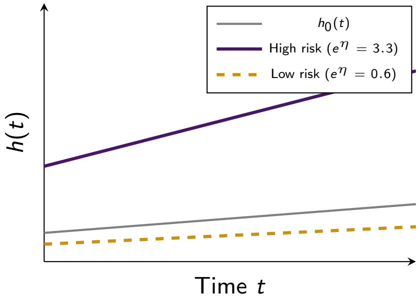
Definition

$$\eta_i = \beta_1 x_{1i} + \cdots + \beta_p x_{pi} = \boldsymbol{\beta}'\mathbf{x}_i$$

Summarises all covariate information for individual i into one number.

How to interpret the risk score:

- Higher $\eta_i \Rightarrow$ higher hazard $\Rightarrow$ shorter expected survival
- $h_i(t) = e^{\eta_i} h_0(t)$: individual i's hazard is e^{η_i} times the baseline
- Two individuals with $\eta_1 > \eta_2$ have a constant hazard ratio $e^{\eta_1 - \eta_2}$ at every time



Why Must the Hazard Be Non-Negative?

The hazard $h(t)$ is a *rate* — it must always be ≥ 0.

The problem with a linear model:

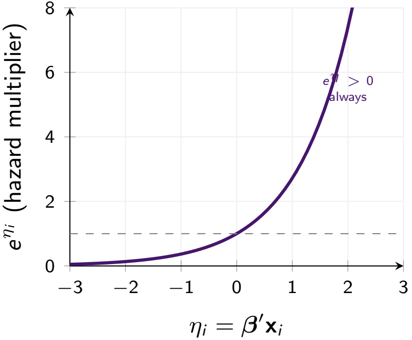
$$h_i(t) = (\beta_1 x_{1i} + \cdots + \beta_p x_{pi})\, h_0(t)$$

For some combinations of large negative β_j and large x_{ji}, the linear predictor could go *negative* $\Rightarrow$ negative hazard. Impossible!

The solution — exponential link:

$$h_i(t) = e^{\eta_i}\, h_0(t), \quad \eta_i \in (-\infty, +\infty)$$

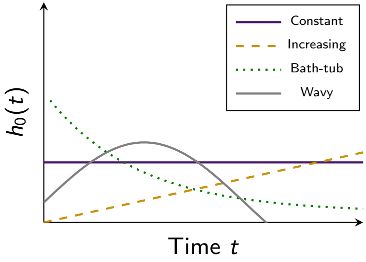
e^{η_i} is *always* positive, regardless of the value of η_i.



The Baseline Hazard — Why We Don't Need to Specify It

$h_0(t)$ = hazard for an individual with *all* $x_j = 0$.

Its shape is **completely free** — any positive function.



Why this is powerful:

- To estimate β (the covariate effects), we do **not** need to know $h_0(t)$
- The partial likelihood (Section 4) eliminates $h_0(t)$ automatically
- This makes the Cox model *robust*: it works regardless of the true shape of the baseline hazard
- We only estimate $h_0(t)$ if we need $\hat{S}_i(t)$ for prediction (Section 11)

Semi-parametric = the best of both worlds: Flexibility of non-parametric methods + power of regression.

What Goes Inside the Cox Model?

The linear predictor $\eta_i = \boldsymbol{\beta}'\mathbf{x}_i$ can contain three types of terms:

Variates	**Factors**	**Interactions / Mixed**
Continuous, e.g. age, Hb, blood pressure	Categorical, e.g. sex, age group, treatment	Combinations of the above
e^{β} per unit increase	e^{α_j} per level	effect depends on both

The construction of the linear predictor follows the same rules as in ordinary linear regression — but the *response* here is the log-hazard ratio, not a mean.

Including a Variate (Continuous Variable)

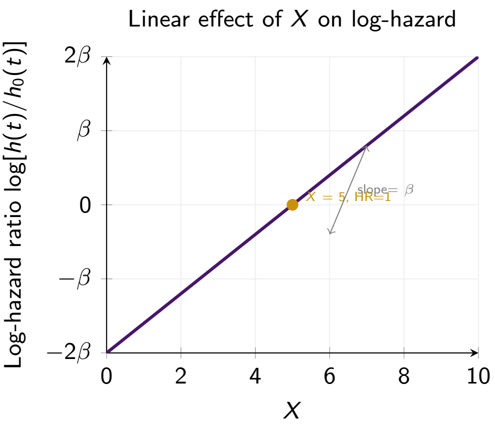
Suppose the hazard depends on one continuous variable X (e.g. serum haemoglobin):

$$h_i(t) = e^{\beta x_i}\, h_0(t)$$

Interpretation:

- Each unit increase in X multiplies the hazard by e^{β}
- A 1-unit increase from $x = 5$ gives the same HR as from $x = 10$
- This is the **linearity assumption**: equal increments give equal multiplicative changes

For r units increase: $\text{HR} = e^{r\beta}$, $\text{SE} = r \cdot \widehat{\text{se}}(\hat{\beta})$.



Including a Factor (Categorical Variable)

A factor with a levels is fitted using $a-1$ **indicator (dummy) variables**, with the first level as *baseline* (effect $=0$).

Age group with 3 levels

Level	A_2	A_3
<60 (baseline)	0	0
60–70	1	0
>70	0	1

$$h_i(t) = e^{\alpha_2 A_{2i} + \alpha_3 A_{3i}} h_0(t)$$

e^{α_2} = HR for 60–70 vs. <60
e^{α_3} = HR for >70 vs. <60

Why $a-1$ variables and not a?

If we defined one indicator per level, we would have a intercepts and no baseline — **overparameterised**.

Setting one level to zero (usually the first or a natural reference group) fixes this.

Which level to pick as baseline?

- The level with most observations
- A natural reference (e.g. "no treatment", youngest age group)
- Whatever makes the HRs easiest to interpret

Two factors A and B **interact** if the effect of A depends on the level of B.

Example: if tumour grade affects hazard differently for men vs. women, sex and grade interact. A model without interaction forces the same grade effect for both sexes — may be wrong.

How to fit an interaction:

Define products of indicator variables: $U_j \times V_k$.

If A has a levels and B has b levels:
Main effects: $(a-1)+(b-1)$ parameters
Interaction: $(a-1)(b-1)$ parameters

Hierarchic principle

Never fit an interaction without the corresponding **main effects**.

A model with $A \times B$ but without A or B is uninterpretable.

Mixed Terms: Factor $\times$ Variate

When is a mixed term needed?

The coefficient of a variate X differs by level of a factor A.
Example: the effect of body weight on risk differs by parity group (number of pregnancies).

How to fit:

Define products $U_j \times X$ for each level j of A, and include X itself:

$$\eta_i = \beta x_i + \alpha_2'(u_{2i}x_i) + \alpha_3'(u_{3i}x_i)$$

Coefficient of X at baseline level: β
Coefficient of X at level 2: $\beta + \alpha_2'$
Coefficient of X at level 3: $\beta + \alpha_3'$

Critical: always include X itself

If you omit X and only include the products U_jX, then for individuals at the baseline level of A (where all $U_j = 0$), the coefficient of X is **zero** — meaning X has no effect for those individuals.

That is almost never intended!

The Challenge: An Unknown $h_0(t)$

To estimate β, we need a likelihood. But **standard MLE** requires the joint probability of the data as a function of all unknowns: β *and* the entire function $h_0(t)$.

The Problem

$h_0(t)$ is an infinite-dimensional unknown. We cannot write a standard likelihood.

Cox's (1972) solution — partial likelihood: construct a likelihood that depends only on **the rank order of events**, not on their actual times. This causes $h_0(t)$ to cancel out completely in the conditional probabilities.

Semi-parametric means: estimate β (parametric part) without ever specifying $h_0(t)$ (non-parametric part).

Building the Partial Likelihood: Step by Step

At each ordered death time $t_{(1)} < t_{(2)} < \cdots < t_{(r)}$:

Step 1: Given that someone in risk set $R(t_{(j)})$ dies, what is the probability it is the observed individual?

Step 2:

$$P\big(\text{ind. } (j) \text{ dies} \mid \text{one death at } t_{(j)}\big) = \frac{e^{\boldsymbol{\beta}'\mathbf{x}_{(j)}}}{\sum_{l \in R(t_{(j)})} e^{\boldsymbol{\beta}'\mathbf{x}_l}}$$

$h_0(t_{(j)})$ appears in numerator *and* denominator and **cancels**!

Step 3: Multiply over all r death times:

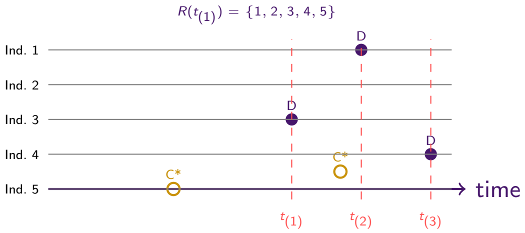
$$L(\boldsymbol{\beta}) = \prod_{j=1}^{r} \frac{e^{\boldsymbol{\beta}'\mathbf{x}_{(j)}}}{\sum_{l \in R(t_{(j)})} e^{\boldsymbol{\beta}'\mathbf{x}_l}}$$

Understanding the Risk Set

$R(t_{(j)})$ = individuals **alive and uncensored** just before time $t_{(j)}$.

Who is in the risk set?

- Everyone who has not yet died
- Everyone who has not yet been censored
- Individuals censored at time $c > t_{(j)}$ are *still in* the risk set at $t_{(j)}$
- Individuals censored at $c \leq t_{(j)}$ are *no longer in* the risk set



Key point

Censored individuals contribute to denominators at all death times before their censoring time — they carry real information!

The Partial Likelihood: Full Worked Example

5 individuals. Write $\psi(i) = e^{\boldsymbol{\beta}'\mathbf{x}_i}$.

Deaths: ind. 3 at $t_{(1)}$, ind. 1 at $t_{(2)}$, ind. 4 at $t_{(3)}$.
Censored: ind. 5 (early), ind. 2 (late).

Death time	Who dies	Risk set
$t_{(1)}$	Ind. 3	$\{1,2,3,4,5\}$
$t_{(2)}$	Ind. 1	$\{1,2,4\}$
$t_{(3)}$	Ind. 4	$\{4\}$

Ind. 5 censored before $t_{(2)}$ $\Rightarrow$ exits. Ind. 2 censored after $t_{(2)}$ $\Rightarrow$ still in $R(t_{(2)})$.

Partial likelihood:

$$L = \frac{\psi(3)}{\psi(1)+\psi(2)+\psi(3)+\psi(4)+\psi(5)}$$

$$\times \frac{\psi(1)}{\psi(1)+\psi(2)+\psi(4)} \times \frac{\psi(4)}{\psi(4)} = 1$$

Third factor $= 1$ trivially. **Log-likelihood:**

$$\log L = \boldsymbol{\beta}'\mathbf{x}_{(3)} - \log \sum_{l=1}^{5} e^{\boldsymbol{\beta}'\mathbf{x}_l} + \boldsymbol{\beta}'\mathbf{x}_{(1)} - \log \sum_{l \in R(t_{(2)})} e^{\boldsymbol{\beta}'\mathbf{x}_l}$$

Why “Partial” Likelihood?

The partial likelihood is **not a true likelihood** because it does not use:

- The actual death times (only their rank order)
- Information from intervals between deaths

Why is this OK?
Between death times, $h_0(t)$ could be anything. Since $h_0(t)$ is unspecified, those intervals carry *no information* about β.

The partial likelihood captures *all available information* about β.

Good news

Standard maximum likelihood theory applies to $L(\beta)$:

- $\hat{\beta}$ is consistent and asymptotically normal
- Variance-covariance matrix from second derivatives
- Likelihood ratio tests are valid
- Wald tests are valid

In R: `coxph()` from the `survival` package maximises $L(\beta)$ automatically using the Newton-Raphson algorithm.

Handling Tied Death Times

The Cox model assumes a **continuous** hazard, so ties shouldn't occur in theory. In practice, times are rounded (days, months) and ties are common.

Breslow (1974)

Treat d_j tied deaths as distinct and sequential.

Default in R (`ties="breslow"`).

Adequate when ties are rare.

Efron (1977)

Better approximation of the exact likelihood.

More computation; usually very close to Breslow.

`ties="efron"`

Cox discrete (1972)

Based on a discrete-time proportional hazards model where ties are natural.

Exact. Slowest.

`ties="exact"`

- When $d_j = 1$ (no ties), all three methods give *identical* results
- For most datasets, Breslow and Efron agree closely

Use Breslow (the default). Switch to Efron if there are many ties at the same time.

Newton-Raphson: How the Model Is Fitted

$\log L(\boldsymbol{\beta})$ has no closed-form solution. Starting from $\hat{\boldsymbol{\beta}}_0 = \mathbf{0}$, iterate:

Newton-Raphson update

$$\hat{\boldsymbol{\beta}}_{s+1} = \hat{\boldsymbol{\beta}}_s + \mathbf{I}^{-1}(\hat{\boldsymbol{\beta}}_s)\,\mathbf{u}(\hat{\boldsymbol{\beta}}_s)$$

$\mathbf{u}$ = gradient; $\mathbf{I}$ = negative Hessian. Stop when changes fall below a tolerance.

Output:

- $\hat{\boldsymbol{\beta}}$: estimated coefficients
- $\mathbf{I}^{-1}(\hat{\boldsymbol{\beta}})$: variance-covariance matrix
- $\widehat{\mathrm{se}}(\hat{\beta}_j) = \sqrt{[\mathbf{I}^{-1}(\hat{\boldsymbol{\beta}})]_{jj}}$

`coxph()` in R does this automatically.

Understanding the algorithm helps you:

- Recognise convergence warnings
- Diagnose collinearity issues
- Understand why overfitting causes huge standard errors

Reading Cox Model Output from R

After `coxph(Surv(time, status) ~ Hb + Bun, data=myeloma)`, R gives:

Variable	$\hat{\beta}$	se	$z = \hat{\beta}/\text{se}$	P (Wald)	$e^{\hat{\beta}}$
Hb	-0.135	0.069	-1.96	0.051	0.874
Bun	0.021	0.006	3.50	0.001	1.021

Wald test for $H_0 : \beta_j = 0$:

$$z = \frac{\hat{\beta}_j}{\widehat{\text{se}}(\hat{\beta}_j)} \sim N(0, 1)$$

Tests whether X_j is needed *given all other variables in the model*.

Interpretation:

- Hb: $z = -1.96$, $P = 0.051$. Borderline significant.
- Bun: $z = 3.50$, $P = 0.001$. Strongly significant.

But remember: these are *conditional* tests. The result for Hb could change if Bun were removed from the model.

Confidence Intervals for β and the Hazard Ratio

Confidence intervals

95% CI for β_j: $\hat{\beta}_j \pm 1.96\,\widehat{\mathrm{se}}(\hat{\beta}_j)$

95% CI for HR e^{β_j}:

$$\left(e^{\hat{\beta}_j - 1.96\,\mathrm{se}_j},\quad e^{\hat{\beta}_j + 1.96\,\mathrm{se}_j}\right)$$

Why exponentiate? $\log\hat{\psi}$ is closer to normal than $\hat{\psi}$, and the log-based CI always stays positive.

Example 3.1 — Breast Cancer

$\hat{\beta} = 0.908$, se $= 0.501$

95% CI for β:
$0.908 \pm 1.96(0.501) = (-0.074, 1.890)$

95% CI for HR:
$(e^{-0.074}, e^{1.890}) = (0.93, 6.62)$

The CI barely includes 1.
$\Rightarrow$ some evidence that HPA staining affects survival, but not conclusive at 5%.

Why Wald Tests Are Not Enough for Model Building

When covariates are correlated, $\hat{\beta}_1, \hat{\beta}_2, \ldots$ are **not independent**.

Illustration: model contains X_1, X_2, X_3. Wald tests say $\hat{\beta}_1$ and $\hat{\beta}_2$ are not significant.

Wrong conclusion: "Only X_3 is needed."

Why wrong: Remove X_2 and the estimate of β_1 may change dramatically — it may become highly significant. The Wald test for $\hat{\beta}_1$ was non-significant only *in the presence of* X_2.

Better approach: compare $-2 \log \hat{L}$ for the model with X_1 vs. without X_1, all else equal.

This **likelihood ratio test** directly measures whether X_1 improves the model after accounting for all other variables.

Wald tests: rough guide only. LRT: the **gold standard** for model comparison.

Questions?

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